

Track-Specific Week 5 Output

Tinuola Owolabi

AnalystLab Africa — Experience Lab | HealthConnect Clinic Project | Data Analytics Track

1. Review of Week 4 Foundation

- Problem defined: HealthConnect Clinic is experiencing a high rate of missed appointments (no-shows) and needs to understand the factors driving this behavior.
- Week 4 approach: Group patients by key factors (distance, reminder status, waiting time, previous no-show history) and compare no-show rates across groups.
- Resources used: HealthConnect Appointment Dataset (5,000 records, 18 columns) and the Data Dictionary.
- Key assumptions/limitations: Missing values in Distance_to_Clinic_KM and Waiting_Time_Minutes; dataset is fictional; linking waiting time to a patient's next visit requires additional data preparation.
- Proposed Week 5 focus: Clean identified data quality issues, calculate proposed KPIs, and begin comparative analysis.

2. Data Preparation

The dataset (5,000 rows, 18 columns) was reviewed for data types, missing values, duplicates, and inconsistent values.

- Data types: All numeric and categorical columns are correctly typed, except booking_date and appointment_date, which are stored as text and were addressed for time-based analysis.
- Missing values: distance_to_clinic_km (90), waiting_time_minutes (60), and reminder_channel (1,366 — expected, since it is blank whenever no reminder was sent).
- Duplicates: None found.
- Inconsistent values: None found — all categorical columns (gender, age_group, appointment_type, appointment_day, appointment_time, reminder_sent, reminder_channel, appointment_outcome) are clean and consistently labeled.

The original dataset remains unaltered; derived columns used for analysis were created and documented separately within the Power BI data model.

3. Exploratory Data Analysis

Q1: Does distance to the clinic affect the likelihood of a no-show?

- Finding: No-show rate increases steadily with distance from the clinic: 46.5% (0-5km), 46.5% (5-10km), 48.5% (10-15km), 51.4% (15-20km), and 57.8% (20km+). Patients living 20km or more from the clinic are noticeably more likely to miss appointments, there is an 11-point gap between the closest and farthest groups.
- Recommendation: Offer telehealth/virtual consultation options for patients living 15km+ from the clinic and prioritize reminder outreach for this group.

Q2: Does receiving an appointment reminder reduce the likelihood of a no-show?

- Finding: Reminders show only a small effect — 47.4%, no-show rate with a reminder vs. 51.4% without, a 4-point difference. Reminders alone do not meaningfully reduce no-shows.
- Recommendation: Keep reminders as a low-cost measure, but pair them with stronger interventions like targeting high-distance or high-risk patients.

Q3: Does long waiting time during a previous visit affect whether a patient attends their next appointment?

- Finding: No-show rate on the next visit is nearly identical regardless of how long the patient waited previously: 49.1% (short wait), 47.1% (medium wait), 49.0% (long wait). Waiting time has no meaningful effect on future attendance.
- Recommendation: Do not prioritize reducing waiting times as a no-show prevention strategy; focus resources on distance and prior no-show history instead.

Q4: Does a patient's history of previous no-shows predict future no-shows?

- Finding: No-show rate rises sharply with a patient's prior no-show count: 43.5% (0 previous), 54.8% (1-2 previous), 68.8% (3+ previous). This is the strongest predictor in the dataset.
- Recommendation: Flag patients with 2+ previous no-shows as high-risk and prioritize them for personal follow-up before their next appointment.

Q5: Are certain patient demographics (gender) associated with different no-show rates?

- Finding: No meaningful difference in no-show rate by gender - Female 48.4%, Male 48.7%, Prefer not to say 43.5% (small sample). Gender does not appear to be a factor in appointment attendance.
- Recommendation: No gender-specific intervention needed; focus resources on factors that show real effects (previous no-show history and distance).

4. KPI Development

1. No-Show Rate

- Definition: Percentage of total appointments resulting in a no-show.
- Why it matters: baseline metric for the entire problem.
- Result: 48.5%.
- Linked to overall problem.
- Interpretation: nearly half of all appointments are missed, confirming a significant operational issue.

2. Total Appointments

- Definition: Total number of appointments in the dataset.
- Why it matters: provides scale/context for interpreting other KPIs.
- Result: 5,000.
- Linked to all business questions.
- Interpretation: a large enough sample to draw reliable conclusions from.

3. Average Distance to Clinic

- Definition: Average distance (km) patients travel to the clinic.
- Why it matters: tests whether distance is a barrier to attendance.
- Result: 10.11 km overall (9.67 km for attended vs. 10.53 km for no-shows).
- Linked to: Question 1.
- Interpretation: no-show patients live measurably farther from the clinic on average.

4. Average Waiting Time

- Definition: Average time (minutes) patients wait at appointments.
- Why it matters: tests whether wait times affect attendance.
- Result: 24.19 minutes overall (24.29 for attended vs. 24.20 for no-shows on the same visit).
- Linked to: Question 3.
- Interpretation: negligible difference... waiting time does not distinguish attendees from no-shows.

5. Total Patients with Previous No-Shows

- Definition: Number of unique patients with at least one prior no-show.
- Why it matters: identifies the scale of repeat-offender risk.
- Result: 1,232 out of 1,696 patients (72.6%).
- Linked to: Question 4.
- Interpretation: missed appointments are a widespread pattern, not isolated to a small few.

5. Dashboard / Visualization

An interactive Power BI dashboard, titled "HealthConnect Patient Attendance Dashboard," was built containing 5 KPI cards (No-Show Rate, Total Appointments, Average Distance, Average Waiting Time, Patients with Previous No-Shows) and 5 charts answering each business question: No-Show Rate by Distance Range (clustered column chart), No-Show Rate by Reminder Status (clustered bar chart), Next Visit No-Show Rate by Waiting Time Bucket (line chart), No-Show Rate by Previous No-Show History (clustered column chart), and No-Show Rate by Gender (donut chart). An Appointment Type slicer allows filtering across all visuals.

6. Business Insights

- Previous no-show history is the strongest predictor of future no-shows, with rates climbing from 43.5% to 68.8% as prior no-shows increase — this is the clinic's clearest opportunity for targeted intervention.
- Distance to the clinic is a genuine, moderate barrier to attendance, with no-show rates rising from 46.5% to 57.8% as distance increases.
- Appointment reminders have only a modest effect (4-point difference) and should not be relied upon as a standalone solution.
- Waiting time does not meaningfully influence whether a patient returns for their next appointment, despite being a common assumption.

- Gender shows no meaningful relationship with no-show behavior, meaning demographic-based targeting by gender is not a useful strategy.
- Overall, nearly half of all appointments (48.5%) result in a no-show, confirming this is a significant, clinic-wide operational issue rather than a minor concern.

7. Cross-Track Collaboration

- Track interacted with: Data Science.
- Information exchanged: The Data Analytics findings particularly that previous no-show history and distance to the clinic are the strongest predictors of no-shows identified as directly relevant inputs for the Data Science track's proposed no-show prediction model.
- Why relevant: These findings help validate which features (`previous_no_shows`, `distance_to_clinic_km`) are likely to carry real predictive signal, potentially saving the Data Science track time during feature selection.
- What changed/improved: The Data Science track can prioritize these two variables as strong candidate features for their baseline model, rather than treating all variables as equally likely to matter.

8. Assumptions, Limitations, Risks and Dependencies (Updated)

- Still relevant: The dataset remains fictional, so findings may not fully reflect real-world patient behavior.
- Resolved: Missing values in `distance_to_clinic_km` and `waiting_time_minutes` were reviewed and did not materially affect the analysis, as they represented a small proportion of records.
- New issue discovered: Building the derived "Next Appointment Outcome" column in Power BI required additional care, as an initial attempt created a row-level column rather than a group-level measure, producing incorrect aggregated percentages until corrected.
- Potential impact: Analytical results depending on derived, multi-step calculations (like Q3) carry a higher risk of technical error and should be independently verified, as was done here using a secondary calculation method.
- How addressed: The Next Visit No-Show Rate was rebuilt as a proper DAX measure and cross-checked against an independent calculation before being finalized in the report.